

Manual

Inspection Planning based on CAD/FMEA EIS-CFM 8.1

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1 Interface for Characteristics CAD/FMEA

Purpose

You use this product to create inspection plan characteristics using a data file and different configuration files. The data file is created using a CAD drawing or a HYDRA-FMEA.

Implementation notes

To import inspection plan characteristics, you require the configuration files. Using the configuration files, you can assign data of inspection plan characteristics that is not included in a data file. You use conditions to flexibly assign the data of inspection plan characteristics.

To create inspection plan characteristics on the MOC, you additionally require the license FEP-PCF or WEP-PCF.

Integration

This product is a complementary component of the HYDRA inspection planning. Using the product "FMEA-PPL", you can import inspection plan characteristics that have been exported from HYDRA-FMEA.

Features

The following functions are available:

- You can create any number of configuration files (XML) to map different import scenarios. Depending on the scenario, different fields of the inspection plan characteristics are pre-populated.
- You can define the fields of inspection plan characteristics if the data of these fields is not included in the CAD drawing or HYDRA-FMEA, but you want to specify a default value for these fields.
- You can automatically generate the OP sequence number.
- You can transfer data of inspection plan characteristics based on a data file with a specified structure. You can process different file types.
- You can add inspection plan characteristics to an existing inspection plan using a specific import configuration.
- You assign the "column" of the data file to a field of the inspection plan characteristic.
- You make rules/specifications to transform column contents of the CAD data file into the format required in HYDRA. This way, you can transfer the previously exported characteristic type "variable" into the format required in HYDRA: "V".
- You can define conditions for the assignment of characteristic data. If you define a condition, the system assigns the control chart XQ with variable characteristics and the p-chart with attributive characteristics, for example.

2 Inspection plan ImportCAD/FMEA - Konfiguration

Purpose

You need to have a configuration file to import inspection plan characteristics based on a CAD drawing or for HYDRA FMEA. The configuration file specifies which data of the data file to be imported is transferred to a specific HYDRA field.

If required, the user can also create their own configuration file.

Following, we outline the functions of a configuration file.

General

The structure of a data file is the same as in a CSV file. The data file is made up from one or more header followed by a data row. Each data row contains exactly one data record. Each data row is the equivalent to the characteristic to be imported. The individual fields of the data file are separated by a separator. A CSV file is separated by a semicolon. In the configuration file for the data import the separator of the data file can be specified. You can define both the separator and the field delimiter character in the configuration file.

If the separator is part of the characteristic data to be imported, enclose the entire field contents with a field delimiter character. If the field delimiter character is part of the characteristic data to be imported, enclose the entire field contents using the field delimiter character. In addition, double the field delimiter character located in the user data.

The following is an example of a data file with a header line, a data line, 2 columns, the separator semicolon, and the field delimiter "double quotation marks" (").

```
column1; column2;
```

```
Value 1;"value containing semicolon; still the same value";
```

Please note/restrictions:

The import process does not support thousands separators for numbers. But you can use a decimal separator a decimal point or a decimal comma. Make sure the decimal separator is not identical to the field delimiter.

The system supports the date values and time stamps in the following format:

```
yyyy-MM-ddTHH:mm:ss[.SSS] (Example: 2018-12-30T13:17:42).
```

The system supports currently only local date values/time stamps in the time zone of the HYDRA WSP service.

The import process interprets a field made up from two field delimiters as a NULL value. The import process also interprets two consecutive separators in a line as a NULL value. Therefore, you cannot create an empty string in the data file. If necessary, you can use a configuration file to convert a NULL value into an empty string.

Structure of a controller configuration file

The configuration file is a JSON file and must contain a value JSON. The configuration file has the FileEncoding UTF-8.

Following, we will explain the basic structure of a configuration file with an example:

```
{
  "Version": "1.0",
  "Settings": {
    "FileType": "CSV",
    "File": {
      "StartRow": 4,
      "ColumnSeparator": ";",
      "QuotingChar": "\"",
      "FileEncoding": "Cp1252"
    }
  },
  "Services": [
    {
      "Name": "InspectionPlanCharacteristic.insert",
      "Parameters": [
        {
          "Name": "inspectionplancharacteristic.op",
          "DataType": "INTEGER",
          "FileColumnNumber": 1
        }
      ],
      "ObjectIdComplexExpression": "Service DummyService.insert called with %sample.acronym%"
    }
  ]
}
```

The characteristic "Version" and "File type" have fixed values.

The characteristic "StartRow" specifies, that the first 3 rows of the data file are header lines and the 4. row contains the first data record.

The characteristic "ColumnSeparator" specifies the separator on a double quotation mark.

The characteristic "QuotingChar" specifies the field delimiter.

The characteristic "FileEncoding" specifies that the FileEncoding to ANSI (Code page Windows 1252). You can look up valid values at the following web address:

<https://docs.oracle.com/javase/8/docs/api/java/nio/charset/Charset.html>

The characteristic "Services" contains the list of all services that the system is supposed to call per data record of the data file. In this example the service "InspectionPlanCharacteristic.insert" is used. This service enables you to create inspection plan characteristics.

The characteristic "Parameter" contains the list of all service parameter. Here, the parameter contains the acronym "inspectionplancharacteristic.op". The acronym "inspectionplancharacteristic.op" contains the OP sequence number of the inspection plan characteristic. The following examples describe how you can identify the acronyms for the individual fields of the inspection plan characteristics.

The characteristic "DataType" specifies the data type. Possible values are "ALPHA_NUMERIC", "INTEGER" and "DECIMAL". In the above example, the data type is "INTEGER", since the OP sequence number is a numeric value without decimal places.

In the example, the characteristics "FileColumnNumber" specifies that the value for the parameter "inspectionplancharacteristic.op" is determined from the first column of the data file when the inspection plan characteristics are imported.

The characteristic „ObjectIdComplexExpression" specifies that the system writes the following text in the log file for each „InspectionPlanCharacteristic.insert" service call:

"Service InspectionPlanCharacteristic.insert called with %inspectionplancharacteristic.op%".

The placeholder "%inspectionplancharacteristic.op%" is replaced by the acronym "inspectionplancharacteristic.op". In addition to the text from "ObjectIdComplexExpression", the log file also contains the status "SUCCESS" or the error that occurred during the service call of "InspectionPlanCharacteristic.insert".

If several parameters are defined in the configuration file, the parameter must be terminated with "},". The last parameter is completed with the final curly bracket (without comma). The end of the parameter definition is completed with "}".

To import inspection plan characteristics, the following parameters must always be specified as follows:

```
"Parameters": [
  {
    "Name": "rectype.id",
    "DataType": "ALPHA_NUMERIC",
    "InitialDataKey": "rectype.id"
  },
  {
    "Name": "area.id",
    "DataType": "ALPHA_NUMERIC",
    "InitialDataKey": "area.id"
  },
  {
    "Name": "inspectionplan.id",
    "DataType": "ALPHA_NUMERIC",
    "InitialDataKey": "inspectionplan.id"
  },
  {
    "Name": "inspectionplan.index",
    "DataType": "ALPHA_NUMERIC",
    "InitialDataKey": "inspectionplan.index"
  },
]
```

The acronyms of the "InitialDataKey" are identified during the import process from the previously marked inspection plan header.

There are two other „InitialDataKey" acronyms to automatically identify and assign the OP sequence number.


```
{
  "Name": "$tmp_var$.startCharacteristicOp",
  "DataType": "INTEGER",
  "InitialDataKey": "startCharacteristicOp"
},
{
  "Name": "$tmp_var$.currentCounter",
  "DataType": "INTEGER",
  "InitialDataKey": "currentCounter"
},
}
```

The parameter has „startCharacteristicOp“ as a fixed value. The "currentCounter" parameter corresponds to the respective data record row of the characteristic to be created in the CSV data file. The first characteristic has the value "1", the second characteristic the value "2", etc.

If you want to assign a fixed parameter, then you need the following parameter definition:

```
{
  "Name": "gmcharacteristic.inspection_mandatory_flag",
  "DataType": "ALPHA_NUMERIC",
  "Value": "TRUE"
},
{
  "Name": "gmcharacteristic.sampling_scheme",
  "DataType": "ALPHA_NUMERIC",
  "Value": "NC"
},
{
  "Name": "gmcharacteristic.sample_size",
  "DataType": "INTEGER",
  "Value": "5"
},
}
```

In the above example, the mandatory test flag is activated, the sampling scheme "n-c" and the sample size "5" are assigned. If the assigned values are based on a status entry, then the ID of the status entry must be assigned as the value.

If the OP sequence number of the inspection plan characteristic is automatically generated, then the following parameter definition is required:

```
{
  "Name": "$tmp_var$.startCharacteristicOp",
  "DataType": "INTEGER",
  "InitialDataKey": "startCharacteristicOp"
},
{
  "Name": "$tmp_var$.currentCounter",
  "DataType": "INTEGER",
  "InitialDataKey": "currentCounter"
},
{
  "Name": "inspectionplancharacteristic.op",
  "DataType": "INTEGER",
  "FormulaValue": "$tmp_var$.startCharacteristicOp + $tmp_var$.currentCounter * 10",
},
}
```

The above example generates an OP sequence number in steps of 10.

If you use a "Dependency" configuration, you can assign fixed values to characteristic contents using a condition. To do so, define a temporary variable first of all. The following example shows a configuration to assign the characteristic type as "variable". The data file to be imported has a defined value as the characteristic type in column "9". In HYDRA, however, the value "V" must be assigned to the inspection plan characteristic. In this case, you have to define a temporary variable for the content to be included to column "9". You then specify that the characteristic type "V" is assigned if the temporary variable contains the content "Variable". If the condition "does not equal" is used, "!=" instead of "==".

```
{
  "Name": "$tmp_var$.inspection.type",
  "DataType": "ALPHA_NUMERIC",
  "FileColumnNumber": 9
},
{
  "Name": "gmcharacteristic.inspection_type.id",
  "DataType": "ALPHA_NUMERIC",
  "Value": "V",
  "Dependency": "$tmp_var$.inspection.type == \"Variable\""
},
}
```

The definition of a condition that is never fulfilled enables to deactivate a parameter definition.

Acronym to import inspection plan characteristics

Open the application "Webservice Data Logging" with the transaction code "syswslog" to specify which acronym corresponds to an inspection plan characteristic field. The recording must be started in the application. Then you have to assign a characteristic to the inspection plan. For this characteristic, you have to assign a value for all fields. Then you have to complete the recording in the application „Webservice Data-Logging“. Check the entry in the list which has content „InspectionCharacteristic.insert“ in the "Service" column. Click on the button "Show details". Pay special attention to the entries with the ending ".param" in the column "Key". You can see the value in the "Value" column of the inspection of the plan characteristic field.

The required acronym corresponds to the entry in the column "Key" without the ending ".param".

The following tables gives you an overview on the existing acronyms to import inspection plan characteristics. The list does not claim to be complete.

area.id
article.designation
article.drawing_number
article.id
article.index
dynamic_modification_standard.aql_value
dynamic_modification_standard.designation
dynamic_modification_standard.id
dynamic_modification_standard.inspection_level.id

dynamic_modification_standard.method.id
fit.id
fit_norm.id
inspectionplan.id
inspectionplan.index
inspectionplancharacteristic.detail_source.id
inspectionplancharacteristic.inspectionplace.id
inspectionplancharacteristic.no_cavity
inspectionplancharacteristic.operation.designation
inspectionplancharacteristic.operation.id
inspectionplancharacteristic.op
inspectionplancharacteristic.qmcharacteristic.machine.group_id
inspectionplancharacteristic.qmcharacteristic.machine_id
inspectionplancharacteristic.qmcharacteristic.machine_planned
inspectionplancharacteristic.sample_group
inspectionplancharacteristic.sampling
inspectionplancharacteristic.specification_source.id
qmanalysisset.id
qmcharacteristic.acceptancequantity
qmcharacteristic.capture_function
qmcharacteristic.capture_param4
qmcharacteristic.capture_param5
qmcharacteristic.capture_percentage
qmcharacteristic.capture_quantity
qmcharacteristic.capture_time_interval
qmcharacteristic.certificates_printing.id
qmcharacteristic.chart1.calculate_action_limit_flag
qmcharacteristic.chart1.calculate_warning_limit_flag
qmcharacteristic.chart1.calculation_action_limit
qmcharacteristic.chart1.calculation_distribution
qmcharacteristic.chart1.calculation_type
qmcharacteristic.chart1.calculation_warning_limit
qmcharacteristic.chart1.chart_id
qmcharacteristic.chart1.limits_preset_base
qmcharacteristic.chart1.lower_action_limit_active_flag
qmcharacteristic.chart1.lower_action_limit
qmcharacteristic.chart1.lower_warning_limit_active_flag
qmcharacteristic.chart1.lower_warning_limit
qmcharacteristic.chart1.meanvalue
qmcharacteristic.chart1.preset_cpk

qmcharacteristic.chart1.preset_mean
qmcharacteristic.chart1.preset_relative_action_limit
qmcharacteristic.chart1.preset_relative_warning_limit
qmcharacteristic.chart1.preset_sigma
qmcharacteristic.chart1.preset_xbar_value
qmcharacteristic.chart1.trend_active_flag
qmcharacteristic.chart1.upper_action_limit_active_flag
qmcharacteristic.chart1.upper_action_limit
qmcharacteristic.chart1.upper_warning_limit_active_flag
qmcharacteristic.chart1.upper_warning_limit
qmcharacteristic.chart2.calculate_action_limit_flag
qmcharacteristic.chart2.calculate_warning_limit_flag
qmcharacteristic.chart2.calculation_action_limit
qmcharacteristic.chart2.calculation_distribution
qmcharacteristic.chart2.calculation_type
qmcharacteristic.chart2.calculation_warning_limit
qmcharacteristic.chart2.chart_id
qmcharacteristic.chart2.limits_preset_base
qmcharacteristic.chart2.lower_action_limit_active_flag
qmcharacteristic.chart2.lower_action_limit
qmcharacteristic.chart2.lower_warning_limit_active_flag
qmcharacteristic.chart2.lower_warning_limit
qmcharacteristic.chart2.meanvalue
qmcharacteristic.chart2.preset_cpk
qmcharacteristic.chart2.preset_mean
qmcharacteristic.chart2.preset_relative_action_limit
qmcharacteristic.chart2.preset_relative_warning_limit
qmcharacteristic.chart2.preset_sigma
qmcharacteristic.chart2.preset_xbar_value
qmcharacteristic.chart2.trend_active_flag
qmcharacteristic.chart2.upper_action_limit_active_flag
qmcharacteristic.chart2.upper_action_limit
qmcharacteristic.chart2.upper_warning_limit_active_flag
qmcharacteristic.chart2.upper_warning_limit
qmcharacteristic.computation_formula_flag
qmcharacteristic.computation_formula
qmcharacteristic.decimal_places
qmcharacteristic.defect_weight
qmcharacteristic.designation
qmcharacteristic.do_inspect_flag

qmcharacteristic.dynamic_modification_type
qmcharacteristic.dynamic_sampling_flag
qmcharacteristic.emu_assignment
qmcharacteristic.emu_characteristic_copy
qmcharacteristic.external_influence
qmcharacteristic.fit_used
qmcharacteristic.gage.family.designation
qmcharacteristic.gage.family.id
qmcharacteristic.gage.id
qmcharacteristic.gage_mode.id
qmcharacteristic.id
qmcharacteristic.incidental_specification_of_parts
qmcharacteristic.initial_inspection_severity.designation
qmcharacteristic.initial_inspection_severity.id
qmcharacteristic.inspection_mandatory_flag
qmcharacteristic.inspection_sequence
qmcharacteristic.inspection_type.id
qmcharacteristic.inspectionresult_base.id
qmcharacteristic.interval_type
qmcharacteristic.interval_unit
qmcharacteristic.interval_value
qmcharacteristic.issuequantity
qmcharacteristic.lower_plausibility_limit_relative
qmcharacteristic.lower_plausibility_limit
qmcharacteristic.lower_process_action_limit_formula
qmcharacteristic.lower_tolerance_limit_active_flag
qmcharacteristic.lower_tolerance_limit_formula
qmcharacteristic.lower_tolerance_limit_relative
qmcharacteristic.lower_tolerance_limit
qmcharacteristic.machine_source_status
qmcharacteristic.machine_status_change_flag
qmcharacteristic.measure_type.id
qmcharacteristic.numberofsamples
qmcharacteristic.only_formula_flag
qmcharacteristic.output_batch_change_flag
qmcharacteristic.recording_detail.id
qmcharacteristic.sample_size
qmcharacteristic.sampling_scheme
qmcharacteristic.shift_change_flag
qmcharacteristic.sigma_process

qmcharacteristic.target_value_formula
qmcharacteristic.target_value
qmcharacteristic.upper_plausibility_limit_relative
qmcharacteristic.upper_plausibility_limit
qmcharacteristic.upper_process_action_limit_formula
qmcharacteristic.upper_tolerance_limit_active_flag
qmcharacteristic.upper_tolerance_limit_formula
qmcharacteristic.upper_tolerance_limit_relative
qmcharacteristic.upper_tolerance_limit
qmcharacteristic.userfield01
qmcharacteristic.visualization_flag
qmcharacteristic.visualization_function
qmcharacteristic.visualization_param4
qmcharacteristic.visualization_param5
qmcharacteristic.visualization_percentage
qmcharacteristic.visualization_position
qmcharacteristic.visualization_quantity
qmcharacteristic.visualization_time_interval
qmcharacteristic.qmdocument.copy_from_masterdata
rectype.id
transitional_definition.designation
transitional_definition.id
unit.id
workplace.commentary
workplace.designation
workplace.id
select.colconf
select.aggfuncts